When Does Policy-Level Permutation Consistency Help Zero-Shot Coordination in Small MARL? A Controlled Study with PICR-MA

Raj Kashikar
Virginia Tech
rajskashikar@vt.edu

Abstract

Teammate features in decentralized multi-agent reinforcement learning are often concatenated in an arbitrary fixed order, even when teammates are exchangeable. We study whether reducing this order sensitivity improves zero-shot coordination with unseen partners. Our testbed is *PICR-MA*, a lightweight policy-side regularizer that samples two independent permutations of teammate feature blocks and penalizes the symmetric Kullback–Leibler divergence between the resulting action distributions. PICR-MA leaves the inference-time policy architecture unchanged and adds only extra training-time forward passes. We evaluate PICR-MA against IPPO, Deep Sets, and Other-Play on PettingZoo MPE (`simple_spread_v3`, `simple_adversary_v3`) and small matrix-game sanity checks using same-family held-out partner cross-play, permutation sensitivity, and runtime diagnostics. In a 5-seed, 50k-step MPE cohort plus a 20k pilot sweep, PICR-MA consistently lowers permutation sensitivity. Its mean same-family held-out cross-play is higher than IPPO on `simple_adversary_v3` but lower on `simple_spread_v3`, and matched-seed comparisons show that the coordination effect is much less stable than the diagnostic change. The main contribution of the paper is therefore not a new default MARL recipe, but a controlled empirical result: policy-level permutation consistency is a useful bias in some cooperative settings, yet invariance diagnostics alone do not predict zero-shot coordination quality.

1 Introduction

Zero-shot coordination (ZSC) asks whether an agent trained in self-play can cooperate with an unfamiliar partner at test time [Hu et al., 2020]. In decentralized MARL, one source of ZSC brittleness is mundane but pervasive: teammate features are commonly fed to a multilayer perceptron in a fixed concatenation order, so the policy may learn distinct roles for the "first" and "second" teammate even when the environment is symmetric. This creates an avoidable dependence on an arbitrary observation convention.

This paper studies a minimal intervention at the policy level. Rather than changing the policy architecture to a set encoder or graph network [Zaheer et al., 2017, Liu et al., 2020], we regularize the policy to produce similar action distributions under different teammate-order views of the same observation. The resulting method, PICR-MA, is intentionally simple: during training we apply two independent permutations to teammate feature blocks and penalize their action-distribution disagreement with a symmetric KL loss.

The interesting question is not only whether this penalty reduces a permutation-sensitivity diagnostic. The more important question is whether that diagnostic reduction improves down-

Preprint.

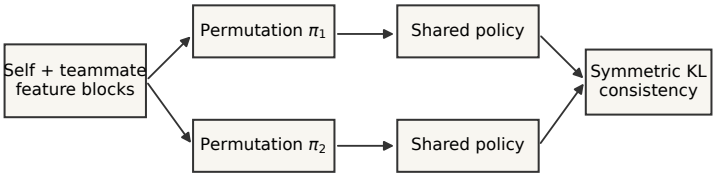


Figure 1: PICR-MA overview. For each training example, the teammate-specific observation blocks are permuted twice, the shared policy is evaluated on both views, and the resulting action distributions are matched with a symmetric KL penalty. Inference uses the original policy with no extra architectural machinery.

stream coordination with held-out partners. The answer in the current artifact is mixed. On `simple_adversary_v3`, PICR-MA raises the cohort-average same-family held-out cross-play score over IPPO while also lowering permutation sensitivity, but the matched-seed gains are inconsistent. On `simple_spread_v3`, it lowers permutation sensitivity and still loses to IPPO, Deep Sets, and Other-Play. This mismatch is the main scientific point of the workshop paper.

Contributions.

- We implement PICR-MA as a policy-side permutation-consistency regularizer for shared-parameter decentralized MARL with no inference-time architectural change.
- We build a controlled evaluation stack with held-out partner cross-play, explicit teammate-block slicing, permutation-sensitivity diagnostics, and runtime tracking on MPE and matrix-game sanity checks.
- We show that PICR-MA reliably reduces permutation sensitivity in the main MPE cohort, but its effect on held-out coordination is environment-dependent: helpful on `simple_adversary_v3` and harmful on `simple_spread_v3`.
- We identify a concrete diagnostic failure mode for this family of ideas: lower teammate-order sensitivity does not by itself imply better zero-shot coordination.

2 Method

Each agent observation is decomposed into non-teammate features s and teammate-specific blocks $U = [u_1, \ldots, u_{n-1}]$. For two independently sampled teammate permutations π_1 and π_2, we form

$$o_{(1)} = [s \| \pi_1(U)], \qquad p_1 = \pi_\theta(\cdot \mid o_{(1)}), \tag{1}$$

$$o_{(2)} = [s \| \pi_2(U)], \qquad p_2 = \pi_\theta(\cdot \mid o_{(2)}). \tag{2}$$

The regularizer is

$$\mathcal{L}_{\text{PICR-MA}} = \lambda \, \mathbb{E}\Big[\tfrac{1}{2}\mathrm{KL}(p_1 \| p_2) + \tfrac{1}{2}\mathrm{KL}(p_2 \| p_1)\Big]. \tag{3}$$

The training objective is $\mathcal{L}_{\text{RL}} + \mathcal{L}_{\text{PICR-MA}}$, where $\mathcal{L}_{\text{RL}}$ is the base on-policy objective. The implementation keeps the base MLP policy unchanged at inference time; the only overhead is the additional training-time forward passes required for the consistency term.

We also implement teammate dropout as an ablation that zeros a random subset of teammate blocks during training. It is not part of the core method and, empirically, it does not strengthen the story.

3 Experimental Setup

Environments. The main evidence comes from two PettingZoo MPE tasks [Terry et al., 2021]: `simple_spread_v3`, a fully cooperative three-agent coverage task, and `simple_adversary_v3`,

Table 1: Main 50k-step, 5-seed MPE results. Lower JS is better. The table makes the workshop framing explicit: PICR-MA lowers order sensitivity, improves IPPO on `simple_adversary_v3`, and loses on `simple_spread_v3`.

Environment	Method	Train return	Cross-play	JS ↓	Env metric
Simple Adversary	IPPO	2.346 ± 1.813	17.429 ± 3.770	0.010 ± 0.007	margin=0.260
Simple Adversary	PICR-MA	6.585 ± 3.502	20.159 ± 2.043	0.005 ± 0.003	margin=0.345
Simple Adversary	Deep Sets	7.965 ± 1.698	11.356 ± 3.244	0.000 ± 0.000	margin=0.392
Simple Adversary	Other-Play	3.603 ± 1.497	20.934 ± 3.429	0.014 ± 0.007	margin=0.310
Simple Spread	IPPO	-21.952 ± 0.472	-25.649 ± 0.271	0.026 ± 0.006	coll=0.164
Simple Spread	PICR-MA	-23.425 ± 0.625	-27.206 ± 0.730	0.013 ± 0.002	coll=0.155
Simple Spread	Deep Sets	-24.776 ± 1.413	-25.550 ± 0.489	0.000 ± 0.000	coll=0.065
Simple Spread	Other-Play	-21.626 ± 0.209	-25.533 ± 0.566	0.025 ± 0.013	coll=0.079

where a cooperative team interacts with an adversary. Matrix coordination and stag-hunt variants are retained as fast sanity checks and are reported in the appendix rather than driving the main claim.

Baselines. We compare four methods under matched training budgets: IPPO, IPPO+PICR-MA, IPPO with a Deep Sets encoder, and Other-Play-style teammate-order randomization [Schulman et al., 2017, Terry et al., 2020, Zaheer et al., 2017, Hu et al., 2020]. The comparison is intentionally practical: Deep Sets provides an architectural invariance baseline, and Other-Play provides a symmetry-randomization baseline with a stronger ZSC pedigree.

Main cohort and pilot ablations. The main MPE cohort uses 5 seeds, 50k environment steps per seed, and held-out partner evaluation from independently trained policies sharing the same run tag. A 20k, 3-seed pilot sweep selected $\lambda \in \{0.02, 0.05, 0.10\}$ for PICR-MA and evaluated teammate dropout at probability 0.1. The main 50k PICR-MA setting uses $\lambda = 0.02$ because it gave the strongest held-out cross-play among the pilot settings, even though larger λ values reduced JS more aggressively.

Evaluation. For each run we record training return, held-out partner cross-play return, permutation sensitivity measured by Jensen–Shannon divergence under teammate-order swaps, and environment-specific diagnostics such as collisions or goal-margin statistics. The headline cross-play number in the main table is same-family held-out cross-play, meaning the ego policy is evaluated against independently trained partners from the same method family and run tag; cross-family partners are reported separately in the appendix. Cross-play results are produced from saved checkpoints with a partner-pool filter that excludes the evaluated run itself and respects the intended run tag.

Implementation and compute. All experiments were executed sequentially on a single Apple Silicon laptop under the repo's lean-safe resource profile. The current learner is a small NumPy actor–critic rather than a full PyTorch PPO stack; this keeps the artifact compact and reproducible, but it also limits external ambition and informs the workshop-scale framing of the paper. IPPO, PICR-MA, and Other-Play share the same MLP backbone and optimization budget, so PICR-MA's extra cost comes from two additional training-time forward passes rather than a larger inference model; only Deep Sets changes capacity (1606 versus 3302 cooperative parameters on `simple_spread_v3`, and 1350 versus 3238 on `simple_adversary_v3`).

4 Results

Permutation consistency is real. Table 1 and Figure 4 show that PICR-MA does what it is supposed to do mechanically. In the main 50k cohort, it lowers teammate-order sensitivity relative to IPPO on both MPE tasks, from 0.0263 ± 0.0063 to 0.0127 ± 0.0019 on `simple_spread_v3` and from 0.0102 ± 0.0066 to 0.0053 ± 0.0025 on `simple_adversary_v3`. Deep Sets drives this diagnostic to zero, as expected for an architectural invariance baseline.

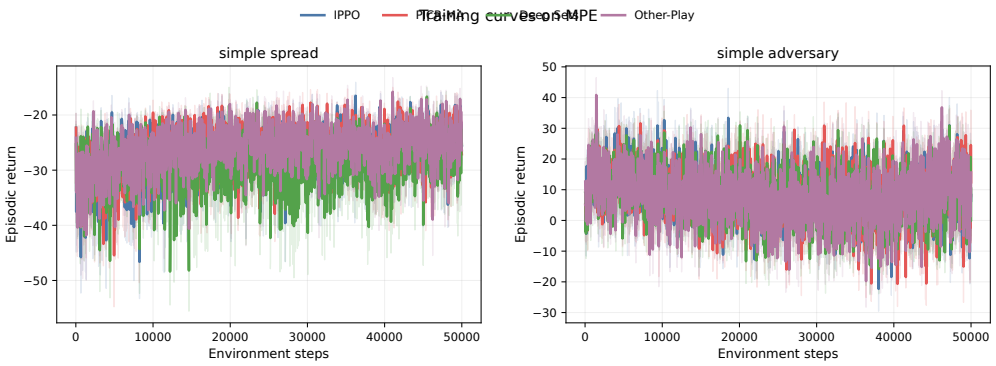


Figure 2: MPE learning curves for the main 50k cohort. The curves show a genuine training signal rather than a toy smoke test, but they also make the scale limitation visible: `simple_spread_v3` is still improving late in training.



Figure 3: Left: same-family held-out partner ZSC summary for the main cohort. Right: matched IPPO→PICR-MA seed-pair deltas. PICR-MA moves permutation sensitivity leftward on both tasks, but the associated cross-play shift is positive only inconsistently on `simple_adversary_v3` and negative in every `simple_spread_v3` pair.

Lower sensitivity does not reliably imply better coordination. Figure 3 sharpens this point with matched IPPO→PICR-MA seed pairs. On `simple_spread_v3`, PICR-MA lowers JS in all five paired comparisons but loses on same-family held-out cross-play in all five; the mean cross-play delta is -1.557 with a 95% bootstrap interval of $[-3.013, -0.651]$. On `simple_adversary_v3`, PICR-MA lowers JS in four of five pairs and improves same-family held-out cross-play in three of five; the mean cross-play delta is $+2.731$ with a wide 95% bootstrap interval of $[-6.141, 12.263]$. The positive adversary result is therefore real at the cohort-average level but much less stable than the diagnostic change, which is exactly the mismatch this paper wants to make concrete.

Baselines matter. The baseline story is also mixed. Deep Sets is the strongest invariance baseline but is weak on `simple_adversary_v3` held-out cross-play, suggesting that exact invariance alone can remove useful asymmetries in that environment. Other-Play is slightly stronger than PICR-MA on `simple_adversary_v3` and clearly stronger on `simple_spread_v3`. The appendix cross-family breakdown also shows that PICR-MA's positive adversary signal is concentrated in same-family partners rather than a uniform gain across partner families. The honest takeaway is therefore not that PICR-MA is a better default than architectural invariance or role randomization, but that it occupies a small simplicity–performance niche: it is lighter than Deep Sets on `simple_spread_v3`, competitive with Other-Play on `simple_adversary_v3`, and simple to bolt onto an existing MLP policy.

Ablations and overhead. The 20k pilot ablations in Table 2 show that stronger regularization does not monotonically help. On `simple_spread_v3`, increasing λ reduces JS but steadily worsens held-out cross-play. On `simple_adversary_v3`, the mild setting $\lambda = 0.02$ gives the best held-out cross-play in the sweep. Teammate dropout harms both tasks. Runtime overhead is moderate rather than negligible: in the main 50k cohort, PICR-MA costs $1.62\times$ IPPO on `simple_spread_v3`

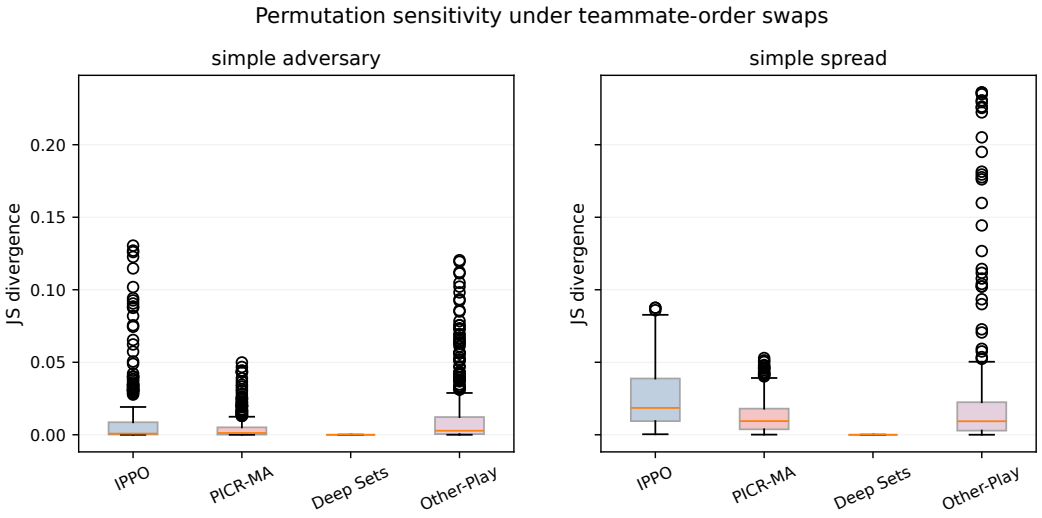


Figure 4: Permutation sensitivity distributions over saved evaluation states. Deep Sets is exactly invariant by construction; PICR-MA improves substantially over IPPO but not always in the way that matters most for held-out coordination.

Table 2: PICR-MA pilot ablations from the 20k, 3-seed sweep. The table supports a narrow claim: mild regularization is preferable, and teammate dropout is not helpful in this artifact.

Environment	Setting	Cross-play	JS ↓	Runtime (s)
Simple Spread	$\lambda = 0.02$	-31.673 $\pm$ 0.520	0.025 $\pm$ 0.006	35.936
Simple Spread	$\lambda = 0.05$	-33.884 $\pm$ 1.281	0.022 $\pm$ 0.004	55.688
Simple Spread	$\lambda = 0.10$	-34.673 $\pm$ 1.060	0.009 $\pm$ 0.002	91.293
Simple Spread	dropout $p = 0.10$	-37.486 $\pm$ 2.504	0.016 $\pm$ 0.004	14.812
Simple Adversary	$\lambda = 0.02$	25.743 $\pm$ 1.344	0.016 $\pm$ 0.006	50.579
Simple Adversary	$\lambda = 0.05$	22.688 $\pm$ 3.005	0.012 $\pm$ 0.002	66.191
Simple Adversary	$\lambda = 0.10$	24.743 $\pm$ 2.973	0.011 $\pm$ 0.003	39.520
Simple Adversary	dropout $p = 0.10$	16.897 $\pm$ 2.086	0.026 $\pm$ 0.016	18.596

and $1.33\times$ IPPO on `simple_adversary_v3`, while remaining well below the Deep Sets cost on `simple_spread_v3`.

5 Discussion

This paper should not be read as evidence that policy-level permutation consistency is a new default recipe for ZSC in MARL. The stronger and more interesting conclusion is narrower: it is possible to reduce order sensitivity substantially with a very small policy-side intervention, but the downstream value of that intervention depends strongly on the environment and the competing inductive biases already present in the baseline.

The contrast between the two MPE tasks suggests a practical hypothesis. In `simple_adversary_v3`, reducing brittle teammate-order dependence appears helpful because the cooperative team benefits from a mild symmetry bias without requiring exact exchangeability. In `simple_spread_v3`, the same bias does not rescue coordination and may suppress distinctions that matter for coverage conventions. The paired-delta analysis makes this concrete: the same intervention moves the diagnostic in the intended direction on both tasks, yet the coordination effect flips sign by environment and does not generalize cleanly across partner families. Invariance diagnostics measure one property of the policy, but they do not summarize the entire coordination problem.

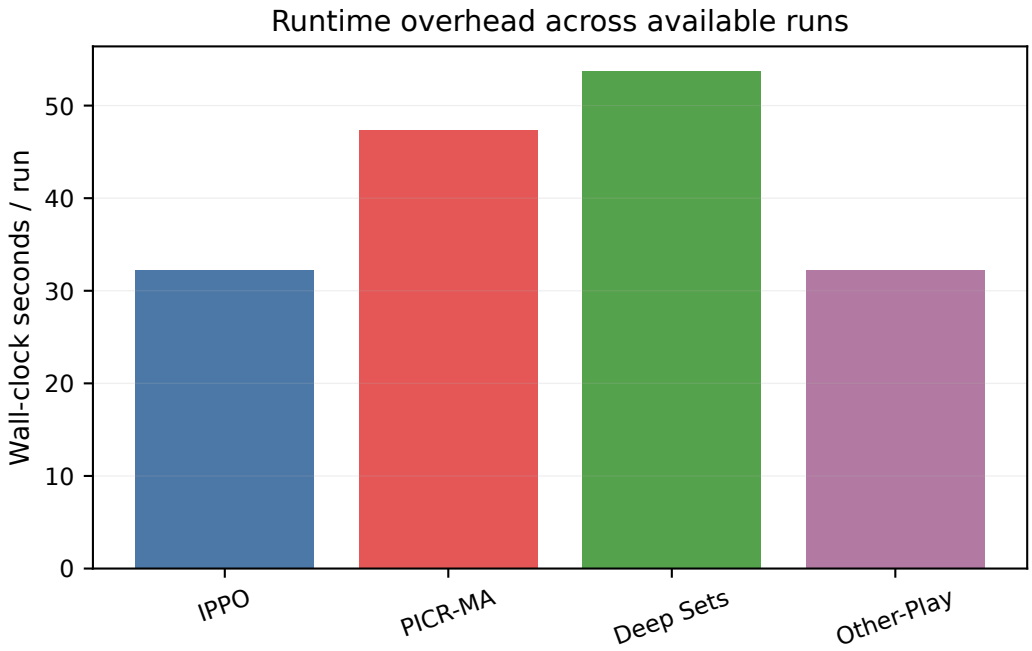


Figure 5: Per-run wall-clock runtime across the main cohort. PICR-MA is meaningfully more expensive than IPPO, but still simpler than an architectural rewrite and cheaper than Deep Sets on `simple_spread_v3`.

Limitations. The learner is a compact NumPy implementation rather than a standard PyTorch PPO codebase. The main MPE cohort is stronger than the original smoke runs, but it remains a 50k-step laptop-scale study rather than a large benchmark campaign. Matrix games are useful sanity checks, not headline evidence. These limitations are acceptable for a focused workshop submission, but they should prevent stronger claims than the ones made here.

6 Conclusion

PICR-MA is a clean way to inject permutation consistency into decentralized MARL without changing the inference-time policy. In the current controlled study, that bias reliably lowers teammate-order sensitivity and raises the mean same-family cross-play score on one MPE task, but the paired-seed gains are unstable and the effect does not transfer uniformly across environments or partner families. For workshop purposes, the more durable contribution is the empirical lesson: lower permutation sensitivity is informative but insufficient as a standalone proxy for zero-shot coordination quality.

Acknowledgments and Disclosure of Funding

This project used only local laptop-scale compute and open-source dependencies. No external experiment tracker, paid API, or cloud compute service was used in the reported results.

References

Hengyuan Hu, Adam Lerer, Alexander Peysakhovich, and Jakob N. Foerster. Other-play for zero-shot coordination. In *Proceedings of the 37th International Conference on Machine Learning*, 2020.

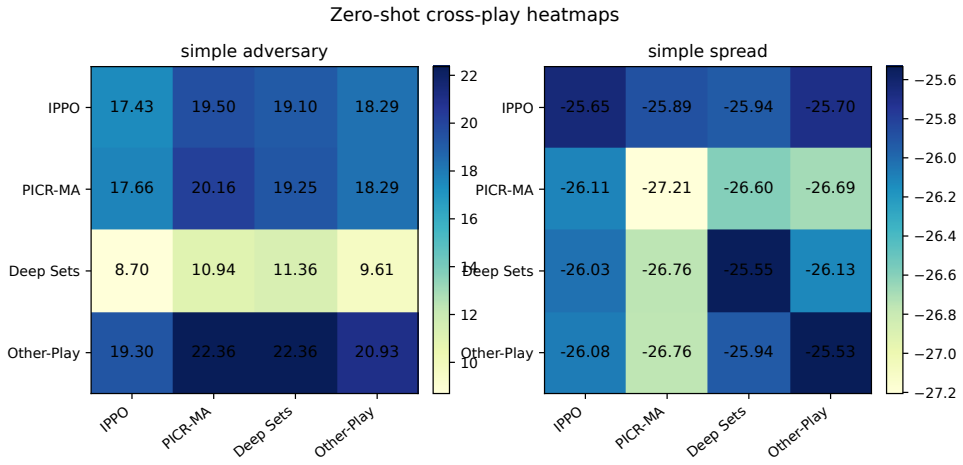


Figure 6: Held-out partner cross-play heatmaps from the main 50k cohort. The main paper uses a compact summary plot; this appendix figure exposes the full family-by-family structure.

Iou-Jen Liu, Raymond A. Yeh, and Alexander G. Schwing. Pic: Permutation invariant critic for multi-agent deep reinforcement learning. In *Proceedings of the 4th Conference on Robot Learning*, 2020.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. arXiv:1707.06347.

Justin K. Terry, Nathaniel Grammel, Sanghyun Son, Benjamin Black, and Anikait Agrawal. Revisiting parameter sharing in multi-agent deep reinforcement learning, 2020. arXiv:2005.13625.

Justin K. Terry, Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis Santos, Rodrigo Perez, Christopher Horsch, Caleb Dieffendahl, Niall L. Williams, Yashas Lokesh, and Praveen Ravi. Pettingzoo: Gym for multi-agent reinforcement learning. In *NeurIPS Datasets and Benchmarks*, 2021.

Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, 2017.

A Additional Cross-Play Results

Figure 6 shows the full cross-play matrices behind the main-paper summary plot. The matrices make two points visible. First, `simple_spread_v3` remains broadly difficult for PICR-MA, not just on the diagonal. Second, on `simple_adversary_v3`, the PICR-MA advantage over IPPO is concentrated in same-family partners rather than a uniform cross-family gain, while Other-Play remains slightly stronger overall.

B Additional Tables

C Matrix-Game Sanity Checks

The matrix games remain useful as fast debugging environments for observation slicing, partner-pool evaluation, and rollout correctness, but they do not determine the paper's main claim. We therefore treat them as supporting evidence rather than headline evidence.

D Observation Slicing, Hyperparameters, and Assets

Observation slicing. Teammate-block indices are explicitly cached per environment rather than inferred implicitly at each update. For `simple_adversary_v3`, the consistency loss is applied only

Table 3: Held-out partner cross-play split into same-family and cross-family partners.

Environment	Ego method	Same-family	Cross-family
Simple Adversary	Deep Sets	11.356 $\pm$ 2.239	9.751 $\pm$ 1.260
Simple Adversary	IPPO	17.429 $\pm$ 2.729	18.964 $\pm$ 1.501
Simple Adversary	Other-Play	20.934 $\pm$ 2.639	21.342 $\pm$ 1.253
Simple Adversary	PICR-MA	20.159 $\pm$ 2.398	18.400 $\pm$ 1.084
Simple Spread	Deep Sets	-25.550 $\pm$ 0.910	-26.305 $\pm$ 0.361
Simple Spread	IPPO	-25.649 $\pm$ 0.584	-25.844 $\pm$ 0.438
Simple Spread	Other-Play	-25.533 $\pm$ 0.782	-26.260 $\pm$ 0.437
Simple Spread	PICR-MA	-27.206 $\pm$ 0.702	-26.466 $\pm$ 0.454

Table 4: Matched-seed PICR-MA versus IPPO deltas. Negative ΔJS is desirable. The table makes the workshop claim explicit: the diagnostic shift is more consistent than the coordination shift.

Environment	Quantity	Mean delta	95% bootstrap CI	Direction count
Simple Spread	Cross-play	-1.557	[-3.013, -0.651]	0/5 higher
Simple Spread	JS $\downarrow$	-0.014	[-0.027, -0.004]	5/5 lower
Simple Adversary	Cross-play	2.731	[-6.141, 12.263]	3/5 higher
Simple Adversary	JS $\downarrow$	-0.005	[-0.015, 0.002]	4/5 lower

to the cooperative team. The precise slicing assumptions and any environment-specific simplifications are recorded in `results/OVERRIDES.md`.

Hyperparameters. The main reported learner uses hidden size 64, learning rate 3×10^{-3}, discount 0.99, GAE parameter 0.95, PPO clip ratio 0.2, and four update epochs per rollout. The main PICR-MA cohort uses $\lambda = 0.02$ and no teammate dropout. Exact commands, profiles, and artifact inventories are listed in `results/FINAL_REPORT.md`.

Model capacity and fairness. IPPO, PICR-MA, and Other-Play share the same inference-time MLP and optimizer settings; only Deep Sets changes the encoder capacity. In the main cohort this corresponds to 1606 cooperative parameters for IPPO/PICR-MA/Other-Play versus 3302 for Deep Sets on `simple_spread_v3`, and 1350 versus 3238 on `simple_adversary_v3`. PICR-MA's extra runtime therefore comes from training-time consistency passes rather than a larger deployed policy.

Assets and licenses. The experiments rely on PettingZoo 1.24.3, Gymnasium 1.2.3, and SuperSuit 3.10.0. Local package metadata reports MIT licenses for these core environment assets. All reported figures and tables are generated from files stored under `results/`.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the paper's scope narrowly and match the empirical evidence: PICR-MA lowers permutation sensitivity, helps on one MPE task, hurts on another, and does not support a universal ZSC claim.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.

Table 5: Permutation sensitivity summary across environments and methods.

Environment	Method	JS divergence
Simple Adversary	Deep Sets	0.000 ± 0.000
Simple Adversary	IPPO	0.010 ± 0.001
Simple Adversary	Other-Play	0.014 ± 0.001
Simple Adversary	PICR-MA	0.005 ± 0.001
Simple Spread	Deep Sets	0.000 ± 0.000
Simple Spread	IPPO	0.026 ± 0.001
Simple Spread	Other-Play	0.025 ± 0.003
Simple Spread	PICR-MA	0.013 ± 0.001

Table 6: Runtime summary relative to IPPO.

Environment	Method	Runtime (s)	Relative to IPPO
Simple Adversary	Deep Sets	44.885	1.34×
Simple Adversary	IPPO	33.504	1.00×
Simple Adversary	Other-Play	32.045	0.96×
Simple Adversary	PICR-MA	44.575	1.33×
Simple Spread	Deep Sets	62.529	2.02×
Simple Spread	IPPO	31.004	1.00×
Simple Spread	Other-Play	32.329	1.04×
Simple Spread	PICR-MA	50.235	1.62×

- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Discussion section explicitly covers the small-compute regime, the NumPy learner, the 50k-step scale limit, and the mixed baseline outcomes.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.

No matrix-game results available

Figure 7: Matrix-game summary under the current small-compute regime.

- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper is empirical and does not present new formal theorems or proofs.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: The paper specifies environments, baselines, seeds, steps, hyperparameters, partner-pool evaluation, and the artifact paths used to generate the figures and tables.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The submission package includes the code, generated results, and scripts used for training, evaluation, aggregation, plots, tables, and manuscript population, together with report files documenting exact commands.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: The main text reports the learner family, environment setup, evaluation protocol, and hyperparameter-selection logic, and the appendix points to exact config files and run reports.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The main-paper figures and tables report mean $\pm$ standard error across seeds or partner samples, and the paper states that the uncertainties are standard errors rather than significance tests.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The paper reports the Apple Silicon laptop setting, the lean-safe resource profile, and per-run wall-clock runtime, and it notes that all experiments were executed sequentially on local compute.

   Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The work is a small-scale empirical study on open-source simulated environments and does not involve deception, human subjects, or restricted data.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The paper includes a broader-impact discussion covering both improved robustness/reproducibility and possible risks from broader deployment of multi-agent systems.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release high-risk generative models, scraped datasets, or similarly dual-use assets.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The paper cites the environment and baseline sources and the appendix records the locally verified MIT license metadata for the core environment packages used in the study.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The repository documents the new code, result artifacts, and manuscript exports with a README, run reports, and explicit reproduction commands.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.

- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The work does not involve crowdsourcing or human-subject data collection.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The reported experiments are entirely in simulation and do not involve human participants.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: LLMs are not part of the method, training pipeline, or experimental claims in this paper.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.